

Space–Time Algebra

A compact development of paravectors and their physical applications

Abstract

This document develops a three-generator Clifford algebra and its complex paravectors, then applies them to relativity, electromagnetism, and quantum mechanics. Natural units $\hbar = c = 1$ and signature $(+, -, -, -)$ are used.

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1 The algebra at a glance

$$\begin{aligned}
\sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\
\mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\
\boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\
(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} + \mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
Z &= (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3.
\end{aligned}$$

Ordinary three-vector operations use

$$\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2 + A_3 B_3, \quad (1.1)$$

$$\mathbf{A} \times \mathbf{B} = \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \quad (1.2)$$

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}, \quad (1.3)$$

$$\|\mathbf{A}\| = \sqrt{\mathbf{A}^2}. \quad (1.4)$$

Thus the square of a vector always denotes its self dot product, not an algebra product. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, define

$$\|Z\|^2 := \|S\|^2 + \|\mathbf{V}\|^2, \quad (1.5)$$

where

$$\|S\|^2 := SS^* = S^* S, \quad (1.6)$$

$$\|\mathbf{V}\|^2 := \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V}. \quad (1.7)$$

On scalar and vector coefficients, $*$ supplies the required complex conjugation; on a full algebra element, $*$ remains sigma conjugation. Thus, for $\mathbf{V} = \mathbf{A} + \mathbf{i}\mathbf{B}$,

$$\|\mathbf{V}\|^2 = \mathbf{A}^2 + \mathbf{B}^2, \quad \mathbf{V}^2 \neq \|\mathbf{V}\|^2 \text{ in general.} \quad (1.8)$$

The commutator is

$$[X, Y] := XY - YX, \quad [Y, X] = -[X, Y]. \quad (1.9)$$

Hence $[X, Y] = 0$ exactly when X and Y commute.

The defining relations in the panel give a graded basis

$$\underbrace{1}_{\text{scalar}}, \quad \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{vectors}}, \quad \underbrace{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3}_{\text{bivectors}}, \quad \underbrace{\mathbf{i}}_{\text{pseudoscalar}}.$$

The pseudoscalar $\mathbf{i} = \sigma_1 \sigma_2 \sigma_3$ commutes with every σ_n , and anticommutation gives $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$.

With the Kronecker delta δ_{nm} and Levi-Civita symbol ϵ_{nmk} ,

$$\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} \mathbf{i}\sigma_k, \quad [\sigma_n, \sigma_m] = 2\epsilon_{nmk} \mathbf{i}\sigma_k. \quad (1.10)$$

Repeated spatial indices are summed. The vector-product identity in the opening panel gives

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (1.11)$$

$$\mathbf{A} \cdot \mathbf{B} = \frac{1}{2}((\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma})). \quad (1.12)$$

2 Involutions and sign conventions

Two involutions are used. Sigma conjugation $(\cdot)^*$ changes the sign of every generator without reversing order. Reversion $(\cdot)^H$ reverses the multiplied order of the sigma generators. In the matrix representation developed in Section 4, this operation is the Hermitian operation; hence the superscript H. Their action by grade is

grade	0	1	2	3
representative	1	σ_n	$i\sigma_n$	i
$(\cdot)^*$	+	-	+	-
$(\cdot)^H$	+	+	-	-
$(\cdot)^{*H}$	+	-	-	+

In particular,

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^H = \sigma_n, \quad i^* = -i, \quad i^H = -i.$$

They obey

$$(ZZ')^H = Z'^H Z^H, \quad (ZZ')^* = Z^* Z'^*, \quad Z^{*H} = Z^{H*}.$$

For a paravector, define the adjugate early for later use:

$$\text{adj}(Z) := Z^{*H}. \quad (2.1)$$

3 Paravectors

3.1 Decompositions

A general complex paravector is

$$Z = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = X + iY = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad (3.1)$$

where $a, b \in \mathbb{R}$, $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$, $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, $S = a + bi$, and $\mathbf{V} = \mathbf{A} + i\mathbf{B}$. The involutions give

$$Z^* = (a - bi) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.2)$$

$$Z^H = (a - bi) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.3)$$

$$Z^{*H} = (a + bi) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.4)$$

Define the four extraction functions by

$$\text{sc}(Z) := S \quad (\text{scalar part}), \quad \text{vec}(Z) := \mathbf{V} \quad (\text{vector coefficient}), \quad (3.5)$$

$$\text{re}(Z) := X \quad (\text{real paravector part}), \quad \text{im}(Z) := Y \quad (\text{imaginary paravector part}). \quad (3.6)$$

The functions re and im also act componentwise on complex scalars and vectors. Thus the components are recovered by

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{*H}), \quad \mathbf{V} = \text{vec}(Z), \quad \mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{*H}), \quad (3.7)$$

$$X = \text{re}(Z) = \frac{1}{2}(Z + Z^H), \quad Y = \text{im}(Z) = \frac{Z - Z^H}{2i}. \quad (3.8)$$

The four real coefficient groups then follow by composition:

$$a = \text{re}(S) = \text{sc}(X) = \text{sc}(\text{re}(Z)), \quad b = \text{im}(S) = \text{sc}(Y) = \text{sc}(\text{im}(Z)), \quad (3.9)$$

$$\mathbf{A} = \text{re}(\mathbf{V}) = \text{vec}(X) = \text{vec}(\text{re}(Z)), \quad \mathbf{B} = \text{im}(\mathbf{V}) = \text{vec}(Y) = \text{vec}(\text{im}(Z)). \quad (3.10)$$

Equivalently, they may be recovered directly from the involutions:

$$a = \frac{1}{4}(Z + Z^* + Z^H + Z^{*H}), \quad (3.11)$$

$$b = \frac{1}{4i}(Z - Z^* - Z^H + Z^{*H}), \quad (3.12)$$

$$\mathbf{A} \cdot \boldsymbol{\sigma} = \frac{1}{4}(Z - Z^* + Z^H - Z^{*H}), \quad (3.13)$$

$$\mathbf{B} \cdot \boldsymbol{\sigma} = \frac{1}{4i}(Z + Z^* - Z^H - Z^{*H}). \quad (3.14)$$

3.2 Products, quadratic forms, and inverse

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$,

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (3.15)$$

Thus

$$\text{sc}(ZZ') = SS' + \mathbf{V} \cdot \mathbf{V}', \quad (3.16)$$

$$\text{vec}(ZZ') = S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}'. \quad (3.17)$$

Equivalently, if $Z = X + iY$ and $Z' = X' + iY'$, then

$$ZZ' = (XX' - YY') + i(XY' + YX').$$

The determinant is

$$\det(Z) = \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V} \cdot \mathbf{V}. \quad (3.18)$$

It may be complex and is not a norm. Whenever $\det(Z) \neq 0$,

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)}. \quad (3.19)$$

For real r and real phase θ , complex scaling gives

$$\det(re^{i\theta}Z) = r^2 e^{2i\theta} \det(Z), \quad (3.20)$$

$$\|re^{-i\theta}Z\|^2 = r^2 \|Z\|^2. \quad (3.21)$$

Every paravector satisfies the quadratic identity

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0. \quad (3.22)$$

Its two algebraic eigenvalues are $\lambda_{\pm} = S \pm \sqrt{\mathbf{V}^2}$.

Vector-product derivation. Using $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \sum_{n,m} A_n B_m \sigma_n \sigma_m \quad (3.23)$$

$$= \sum_{n,m} A_n B_m (\delta_{nm} + \epsilon_{nmk} i \sigma_k) \quad (3.24)$$

$$= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.25)$$

Hence

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^2 = \mathbf{A}^2, \quad (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} = (\mathbf{A}^2)^n, \quad (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} = (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (3.26)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma}) \boldsymbol{\sigma} = \mathbf{A} + i \mathbf{A} \times \boldsymbol{\sigma}, \quad [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] = -2i \mathbf{A} \times \boldsymbol{\sigma}, \quad (3.27)$$

$$[\mathbf{A} \cdot i \boldsymbol{\sigma}, \boldsymbol{\sigma}] = 2 \mathbf{A} \times \boldsymbol{\sigma}, \quad [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.28)$$

3.3 Real products and functions

For $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$,

$$XY = (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.29)$$

$$\text{sc}(XY) = ab + \mathbf{A} \cdot \mathbf{B}, \quad (3.30)$$

$$\text{vec}(XY) = a\mathbf{B} + b\mathbf{A} + i\mathbf{A} \times \mathbf{B}, \quad (3.31)$$

$$[X, Y] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.32)$$

$$X^2 = (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}, \quad (3.33)$$

$$\text{sc}(X^2) = a^2 + \mathbf{A}^2, \quad (3.34)$$

$$\text{vec}(X^2) = 2a\mathbf{A}, \quad (3.35)$$

$$X^2 - 2aX + (a^2 - \mathbf{A}^2) = 0, \quad (3.36)$$

$$X^{-1} = \frac{a - \mathbf{A} \cdot \boldsymbol{\sigma}}{a^2 - \mathbf{A}^2}, \quad (3.37)$$

$$d(X^{-1}) = -X^{-1}(dX)X^{-1}. \quad (3.38)$$

The condition $\mathbf{A} \times \mathbf{B} = 0$ implies $XY = YX$, $X^{-1}Y = YX^{-1}$, and $\det(XY) = \det(Y)\det(X)$. If $X^2 = Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, then

$$a^2 = \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right), \quad \mathbf{A}^2 = b - a^2,$$

with signs and vector direction chosen consistently. Conversely, in the real domain where $b^2 > \mathbf{B}^2$, $b > 0$, and $\mathbf{B} \neq 0$,

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \text{atanh} \left(\frac{\|\mathbf{B}\|}{b} \right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.39)$$

Exponentials. For $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $\mathbf{u} = \mathbf{A}/\|\mathbf{A}\|$,

$$e^X = e^a (\cosh \|\mathbf{A}\| + \sinh \|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (3.40)$$

$$e^{-i\mathbf{A} \cdot \boldsymbol{\sigma}} = \cos \|\mathbf{A}\| - i \sin \|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.41)$$

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $z = \sqrt{\mathbf{V}^2}$,

$$e^Z = e^S \left(\cosh z + \frac{\sinh z}{z} \mathbf{V} \cdot \boldsymbol{\sigma} \right), \quad (3.42)$$

with the quotient defined by continuity at $z = 0$.

3.4 Complex product components

Write $Z = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ and use primes for a second paravector. The four real components of ZZ' are

$$\text{real scalar: } aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \quad (3.43)$$

$$\text{imaginary scalar: } ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \quad (3.44)$$

$$\text{real vector: } a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \quad (3.45)$$

$$\text{imaginary vector: } a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \quad (3.46)$$

For the important quadratic products:

$$Z^2 : (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2i(ab + \mathbf{A} \cdot \mathbf{B}) + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.47)$$

$$ZZ^*, Z^*Z : (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) \mp 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.48)$$

$$ZZ^H, Z^H Z : a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} \pm \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.49)$$

$$Z \text{ adj}(Z) : a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \quad (3.50)$$

The scalar part is cyclic, $\text{sc}(ZZ') = \text{sc}(Z'Z)$, and

$$(ZZ')^* = Z^*Z'^*, \quad (ZZ')^H = Z'^H Z^H, \quad (ZZ')^{*H} = Z'^{*H} Z^{*H}. \quad (3.51)$$

The paired signs follow the displayed product order. Thus $ZZ^* = Z^*Z$ and $ZZ^H = Z^H Z$ precisely when $\mathbf{A} \times \mathbf{B} = 0$.

3.5 Differentiated fields

For products of fields,

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (3.52)$$

Let $F = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ depend on $(t, \mathbf{r})$. Then

$$\partial F = (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \quad (3.53)$$

$$+ (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \quad (3.54)$$

$$+ i(\partial_{\mathbf{r}} b + \partial_{\mathbf{r}} \times \mathbf{A} + \partial_t \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.55)$$

Multiplication by i rotates these four components: the real scalar becomes $-\partial_t b - \partial_{\mathbf{r}} \cdot \mathbf{B}$, the imaginary scalar becomes $\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real vector becomes the negative of the imaginary vector above, and the imaginary vector becomes the real vector.

4 Pauli-matrix representation

The faithful 2×2 complex representation is

$$\text{mat}(\sigma_1) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{mat}(\sigma_2) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{mat}(\sigma_3) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{pmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{pmatrix}. \quad (4.1)$$

The representation preserves identity, addition, and multiplied order. In particular,

$$\text{mat}(1) = \mathbf{I}_2, \quad \text{mat}(Z'Z) = \text{mat}(Z') \text{mat}(Z). \quad (4.2)$$

Let $\mathbf{M} = \text{mat}(Z)$. Four distinct operations occur in the matrix description:

symbol	matrix operation	relation to the algebra
$\mathbf{M}^\top$	transpose	reverses matrix-product order
$\mathbf{M}^\mathbf{C}$	entrywise complex conjugation	sends i to $-i$
$\mathbf{M}^\mathbf{H}$	Hermitian transpose	$\text{mat}(Z^\mathbf{H})$
$\mathbf{M}^*$	sigma conjugation	$\text{mat}(Z^*)$

Thus

$$\mathbf{M}^\mathbf{H} = (\mathbf{M}^\mathbf{C})^\top = (\mathbf{M}^\top)^\mathbf{C}. \quad (4.3)$$

Transpose and Hermitian transpose reverse product order, whereas entrywise complex conjugation and sigma conjugation preserve it:

$$(\mathbf{M}\mathbf{N})^\top = \mathbf{N}^\top \mathbf{M}^\top, \quad (\mathbf{M}\mathbf{N})^\mathbf{C} = \mathbf{M}^\mathbf{C} \mathbf{N}^\mathbf{C}, \quad (4.4)$$

$$(\mathbf{M}\mathbf{N})^\mathbf{H} = \mathbf{N}^\mathbf{H} \mathbf{M}^\mathbf{H}, \quad (\mathbf{M}\mathbf{N})^* = \mathbf{M}^* \mathbf{N}^*. \quad (4.5)$$

In particular,

$$\text{tr}(Z) = 2S, \quad \det(Z) = S^2 - \mathbf{V}^2, \quad \text{mat}(\text{adj}(Z)) = \text{adj}(\mathbf{M}). \quad (4.6)$$

The coefficients and adjugate are

$$S = \frac{1}{2} \text{tr}(\mathbf{M}), \quad V_n = \frac{1}{2} \text{tr}(\sigma_n \mathbf{M}), \quad \text{adj}(\mathbf{M}) = \text{tr}(\mathbf{M}) \mathbf{I}_2 - \mathbf{M}. \quad (4.7)$$

Sigma conjugation is not entrywise complex conjugation. With $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\mathbf{J}^2 = -\mathbf{I}_2$, it is the symplectic conjugation

$$\mathbf{M}^* = -\mathbf{J} \mathbf{M}^\mathbf{C} \mathbf{J}. \quad (4.8)$$

Consequently $\text{mat}(\text{adj}(Z)) = (\mathbf{M}^*)^\mathbf{H} = \text{adj}(\mathbf{M})$. Equation (3.22) is precisely the Cayley–Hamilton theorem for $\mathbf{M} = \text{mat}(Z)$.

5 Space–time kinematics

5.1 Intervals, velocity, and momentum

Represent a differential event by the real paravector

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \quad dX^* = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}. \quad (5.1)$$

The Minkowski interval is

$$ds^2 = \det(dX) = dX \, dX^* = dt^2 - d\mathbf{r}^2. \quad (5.2)$$

It is time-like, null, or space-like according as this expression is positive, zero, or negative. For two intervals,

$$\det(dX + dX') - \det(dX) - \det(dX') = 2(dt \, dt' - d\mathbf{r} \cdot d\mathbf{r}'), \quad [dX, dX'] = 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \quad (5.3)$$

When $ds^2 \neq 0$,

$$dX^{-1} = \frac{dX^*}{ds^2}. \quad (5.4)$$

A null interval has $dt^2 = d\mathbf{r}^2$ and no inverse. Along a time-like worldline, s is proper time and

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \gamma = \frac{dt}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}, \quad U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \det(U) = 1. \quad (5.5)$$

Thus $U^{-1} = U^*$. Proper time obeys $\tau = \int \sqrt{1 - \mathbf{v}^2} \, dt = \int dt/\gamma$. The four-momentum is the paravector

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma\mathbf{v}, \quad E^2 - \mathbf{P}^2 = m^2. \quad (5.6)$$

For coordinate acceleration $\mathbf{a} = d\mathbf{v}/dt$,

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}) \cdot \boldsymbol{\sigma}, \quad (5.7)$$

and the normalization of U implies $\text{sc}(U^* dU/ds) = 0$. If $A = dU/ds$, then

$$\det(A) = -\gamma^4(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2). \quad (5.8)$$

For $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$, define $\mathbf{a}_{\parallel} = (\mathbf{a} \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel}$. Then

$$\det(A) = -\gamma^4(\mathbf{a}_{\perp}^2 + \gamma^2\mathbf{a}_{\parallel}^2). \quad (5.9)$$

The product U^*A has zero scalar part, real vector $\gamma^3\mathbf{a}$, and imaginary vector $-\gamma^3\mathbf{v} \times \mathbf{a}$; while UA has real scalar $2\gamma^5\mathbf{v} \cdot \mathbf{a}$, real vector $\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$, and imaginary vector $\gamma^3\mathbf{v} \times \mathbf{a}$.

5.2 Free-particle action

The free-particle action and Lagrangian are

$$S = -m \int ds = \int L \, dt, \quad L = -m\sqrt{1 - \mathbf{v}^2} = -\frac{m}{\gamma}. \quad (5.10)$$

For a general Lagrangian $L(t, \mathbf{r}, \mathbf{v})$,

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L, \quad H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L, \quad \frac{dH}{dt} = -\partial_t L. \quad (5.11)$$

Thus spatial translation invariance conserves canonical momentum and time translation invariance conserves the Hamiltonian. They yield canonical momentum $\partial_{\mathbf{v}} L = m\gamma \mathbf{v}$ and Hamiltonian $H = m\gamma$. At low velocity,

$$L = -m + \frac{1}{2}m\mathbf{v}^2 + O(v^4), \quad H = m + \frac{1}{2}m\mathbf{v}^2 + O(v^4).$$

6 Electromagnetism

6.1 Field equation and potentials

Introduce the paravector differential operator, electromagnetic field, and source

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (6.1)$$

All four Maxwell equations are the single equation

$$\boxed{\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.} \quad (6.2)$$

Its four components are

$$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (6.3)$$

$$\partial_{\mathbf{r}} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (6.4)$$

Because

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2 =: \square, \quad (6.5)$$

applying ∂^* to (6.2) gives the wave equations and charge conservation:

$$\square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad \square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}, \quad \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0. \quad (6.6)$$

For the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, define

$$S := \text{sc}(\partial \Phi) = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \quad \partial \Phi = S + F^*. \quad (6.7)$$

Hence $\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V$ and $\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}$. Under the gauge transformation

$$\Phi' = \Phi + \partial^* \lambda, \quad (6.8)$$

the field is unchanged and $S' = S + \square \lambda$. The Lorenz gauge is $S = 0$, in which $\rho = \square V$ and $\mathbf{J} = \square \mathbf{A}$. Equivalently, $\text{sc}(\partial \Phi') = S + \square \lambda$ while $\text{vec}(\partial \Phi') = \text{vec}(\partial \Phi)$.

6.2 Invariants, energy, and force

Two simple products contain the standard electromagnetic invariants and energy flow:

$$\frac{1}{2} F^2 = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) + i \mathbf{E} \cdot \mathbf{B}, \quad (6.9)$$

$$\frac{1}{2} F F^H = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (6.10)$$

Thus the scalar and vector parts of the second expression are the field energy density and Poynting vector. The covariant Lorentz force is

$$m \frac{dU}{ds} = q \text{re}(FU) = \frac{q}{2} (FU + UF^H), \quad (6.11)$$

which reproduces $d(m\gamma)/dt = q\mathbf{v} \cdot \mathbf{E}$ and $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.

6.3 Derivation from the potential and action

The potential gradient follows directly from the vector-product identity:

$$\partial\Phi = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A} + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma} + \mathbf{i}(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma} \quad (6.12)$$

$$= S + F^*. \quad (6.13)$$

Consequently

$$F^* = \partial\Phi - S, \quad F = \partial^* \Phi^* - S, \quad \rho = \square V - \partial_t S, \quad \mathbf{J} = \square \mathbf{A} + \partial_{\mathbf{r}} S. \quad (6.14)$$

The interacting particle action is

$$S = -m \int ds - q \int \text{sc}(\Phi^* U) ds \quad (6.15)$$

$$= \int \left(-\frac{m}{\gamma} - q \text{sc}(\Phi^* U / \gamma) \right) dt \quad (6.16)$$

$$= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt. \quad (6.17)$$

Thus

$$L = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad \partial_{\mathbf{v}} L = \mathbf{P} + q\mathbf{A}, \quad H = m\gamma + qV. \quad (6.18)$$

Using $(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A} = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})$ in the Euler–Lagrange equation gives

$$\frac{d\mathbf{P}}{dt} = -q\partial_{\mathbf{r}} V - q\partial_t \mathbf{A} + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}) \quad (6.19)$$

$$= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (6.20)$$

while $dH/dt = -\partial_t L$ gives $d(m\gamma)/dt = q\mathbf{v} \cdot \mathbf{E}$.

To see equation (6.11) componentwise, expand

$$\frac{FU}{\gamma} = \mathbf{v} \cdot \mathbf{E} + \mathbf{i}\mathbf{v} \cdot \mathbf{B} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + \mathbf{i}(\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (6.21)$$

Taking the real paravector part removes the two imaginary components.

6.4 Energy balance

Let

$$T := \frac{1}{2} F F^{\mathbf{H}} = T_s + \mathbf{T}_v \cdot \boldsymbol{\sigma}, \quad T_s = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2), \quad \mathbf{T}_v = \mathbf{E} \times \mathbf{B}. \quad (6.22)$$

Then the scalar part of ∂T is $\partial_t T_s + \partial_{\mathbf{r}} \cdot \mathbf{T}_v$. The source products needed for local balance have components

$$F D^* : -\mathbf{E} \cdot \mathbf{J} - \mathbf{i}\mathbf{B} \cdot \mathbf{J} + (\rho \mathbf{E} + \mathbf{B} \times \mathbf{J}) \cdot \boldsymbol{\sigma} + \mathbf{i}(\rho \mathbf{B} - \mathbf{E} \times \mathbf{J}) \cdot \boldsymbol{\sigma}, \quad (6.23)$$

$$D^* F : -\mathbf{E} \cdot \mathbf{J} - \mathbf{i}\mathbf{B} \cdot \mathbf{J} + (\rho \mathbf{E} - \mathbf{B} \times \mathbf{J}) \cdot \boldsymbol{\sigma} + \mathbf{i}(\rho \mathbf{B} + \mathbf{E} \times \mathbf{J}) \cdot \boldsymbol{\sigma}. \quad (6.24)$$

Since the derivative is not an ordinary scalar derivation,

$$\partial(F^2) = D^* F + F D^* + [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}} F. \quad (6.25)$$

Writing $F = \mathbf{F} \cdot \boldsymbol{\sigma}$, its last term is

$$-2\mathbf{i}\mathbf{F} \cdot (\partial_{\mathbf{r}} \times \mathbf{F}) - 2 \left((\partial_{\mathbf{r}} \cdot \mathbf{F}) \mathbf{F} - \frac{1}{2} \partial_{\mathbf{r}}(F^2) \right) \cdot \boldsymbol{\sigma}. \quad (6.26)$$

7 Relativistic wave equations

7.1 Paravector amplitudes and physical spinors

A paravector amplitude ψ maps to the complex matrix $\mathbf{M}_\psi = \text{mat}(\psi)$. Therefore $\text{mat}(\psi^H \psi) = \mathbf{M}_\psi^H \mathbf{M}_\psi$, which is generally a Hermitian matrix rather than a scalar. Adjugation does not fix this normalization issue: $\text{adj}(\psi) \psi = \det(\psi)$ measures the determinant instead.

The scalar inner product on the full paravector coefficient space is the Hilbert–Schmidt product

$$\langle \phi | \psi \rangle := \text{sc}(\phi^H \psi) = \frac{1}{2} \text{tr}(\mathbf{M}_\phi^H \mathbf{M}_\psi), \quad \langle \psi | \psi \rangle = \|\psi\|^2. \quad (7.1)$$

Thus a full paravector is normalized by $\text{sc}(\psi^H \psi) = 1$, not by $\psi^H \psi = 1$. For an observable $Z^H = Z$ acting from the left, its scalar expectation is

$$\langle Z \rangle_\psi := \frac{\text{sc}(\psi^H Z \psi)}{\text{sc}(\psi^H \psi)}. \quad (7.2)$$

A physical two-component Pauli spinor occupies a minimal left ideal rather than the full matrix algebra. Choose any fixed unit reference direction $\mathbf{u}_0$. In the algebra, define

$$\Pi_+(\mathbf{u}_0) := \frac{1}{2}(1 + \mathbf{u}_0 \cdot \boldsymbol{\sigma}), \quad \psi' := \psi \Pi_+(\mathbf{u}_0). \quad (7.3)$$

In matrix form this is

$$\Pi_+(\mathbf{u}_0) := \frac{1}{2}(\mathbf{I}_2 + \mathbf{u}_0 \cdot \boldsymbol{\sigma}), \quad \mathbf{M}_{\psi'} := \text{mat}(\psi') = \mathbf{M}_\psi \Pi_+(\mathbf{u}_0). \quad (7.4)$$

Let $\mathbf{e}_+(\mathbf{u}_0)$ be a normalized positive eigenvector,

$$(\mathbf{u}_0 \cdot \boldsymbol{\sigma}) \mathbf{e}_+(\mathbf{u}_0) = \mathbf{e}_+(\mathbf{u}_0), \quad \mathbf{e}_+(\mathbf{u}_0)^H \mathbf{e}_+(\mathbf{u}_0) = 1. \quad (7.5)$$

A convenient phase choice, valid for $u_{0,3} \neq -1$, is

$$\mathbf{e}_+(\mathbf{u}_0) = \frac{\{1 + u_{0,3}, u_{0,1} + i u_{0,2}\}^T}{\sqrt{2(1 + u_{0,3})}}; \quad \mathbf{e}_+(-\mathbf{e}_3) = \{0, 1\}^T. \quad (7.6)$$

Define the conventional column spinor $\chi \in \mathbb{C}^2$ by

$$\Pi_+(\mathbf{u}_0) = \mathbf{e}_+(\mathbf{u}_0) \mathbf{e}_+(\mathbf{u}_0)^H, \quad \chi := \mathbf{M}_\psi \mathbf{e}_+(\mathbf{u}_0). \quad (7.7)$$

The algebra, rank-one matrix, and column forms of the state are related by

$$\chi = \mathbf{M}_{\psi'} \mathbf{e}_+(\mathbf{u}_0), \quad \mathbf{M}_{\psi'} = \chi \mathbf{e}_+(\mathbf{u}_0)^H. \quad (7.8)$$

Thus ψ' and χ contain the same two complex components. The matrix $\mathbf{M}_{\psi'}$ has rank one; it has χ as its first column only in a basis whose first vector is $\mathbf{e}_+(\mathbf{u}_0)$. Conversely, $\chi \mathbf{e}_+(\mathbf{u}_0)^H$ uniquely determines ψ' through the faithful matrix representation. Thus χ is a 2×1 vector, not a matrix, and $\chi^H \chi$ is a scalar. The matrix and paravector forms agree:

$$\chi^H \chi = \text{tr}(\mathbf{M}_{\psi'}^H \mathbf{M}_{\psi'}) = 2 \text{sc}(\psi'^H \psi'), \quad (7.9)$$

$$\chi^H \text{mat}(Z) \chi = \text{tr}(\mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_{\psi'}) = 2 \text{sc}(\psi'^H Z \psi'). \quad (7.10)$$

The factor two cancels in normalized expectations. Accordingly, an unrestricted paravector matrix represents two Pauli-spinor columns; matrix right projection by $\mathbf{\Pi}_+$, equivalently algebra right projection by Π_+ , selects one physical spinor. For the Dirac block wavefunction below, applying the same right projection to both components selects one conventional four-component Dirac column.

For a separate transformation $\psi_T := T\psi$, the normalized expectation is

$$\langle Z \rangle_{\psi_T} = \frac{\text{sc}(\psi^H T^H Z T \psi)}{\text{sc}(\psi^H T^H T \psi)}. \quad (7.11)$$

For a unitary rotation the denominator is unchanged. A finite boost is not unitary, so this positive equal-time norm is not itself Lorentz invariant; the covariant Dirac normalization is instead constructed from the conserved Dirac current on a spacelike hypersurface.

7.2 Klein–Gordon equation

With $E = i\partial_t$ and $\mathbf{P} = -i\partial_r$, the mass-shell relation becomes the scalar wave equation

$$(\square + m^2)\Psi = 0. \quad (7.12)$$

Writing $\Omega = \sqrt{m^2 - \partial_r^2}$ gives the frequency decomposition

$$(\partial_t \pm i\Omega)\Psi_{\pm} = 0, \quad \Psi = \Psi_+ + \Psi_-. \quad (7.13)$$

The square-root operator is understood spectrally: on a plane wave with wave vector $\mathbf{k}$, Ω multiplies by $\sqrt{m^2 + \mathbf{k}^2}$.

Minimal electromagnetic coupling replaces the operators by

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_r - q\mathbf{A}, \quad (7.14)$$

so that $(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0$.

7.3 Dirac equation in block form

For a paravector Z , define

$$\mathbf{W}(Z) := \begin{pmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{pmatrix}. \quad (7.15)$$

It satisfies

$$\mathbf{W}(Z)^2 = \text{diag}(Z \text{adj}(Z), \text{adj}(Z) Z), \quad \mathbf{W}(Z)^* = \mathbf{W}(Z^*), \quad \mathbf{W}(Z)^H = \mathbf{W}(Z^H). \quad (7.16)$$

Since $\mathbf{W}(i\partial)^2 = -\square \mathbf{I}$, the first-order equation

$$\boxed{(\mathbf{W}(i\partial) - m\mathbf{I})\Psi = 0} \quad (7.17)$$

squares to the Klein–Gordon equation. For $\Psi = \{\Psi_1, \Psi_2\}^T$ it reads

$$i\partial\Psi_2 - m\Psi_1 = 0, \quad i\partial^*\Psi_1 - m\Psi_2 = 0. \quad (7.18)$$

The characteristic equation is $(s + m)^2 + \square = 0$, hence $s = -m \pm i\sqrt{\square}$. With a potential, the gauge-covariant form is

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0. \quad (7.19)$$

It is invariant under $\Phi' = \Phi + \partial^*\lambda$ and $\Psi = e^{iq\lambda}\Psi'$.

7.4 Schrödinger–Pauli limit

Removing the rest-energy oscillation and expanding at low momentum gives the Pauli equation

$$\mathrm{i}\partial_t\Psi = \left(\frac{(-\mathrm{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right) \Psi. \quad (7.20)$$

For a unit direction $\mathbf{u}$, the spin projectors

$$\Pi_{\pm}(\mathbf{u}) = \frac{1}{2}(\mathbf{I}_2 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) \quad (7.21)$$

satisfy $\Pi_{\pm}^2 = \Pi_{\pm}$, $\Pi_+\Pi_- = 0$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi_{\pm} = \pm\Pi_{\pm}$. Thus $\Psi = \Psi_+ + \Psi_-$ with $\Psi_{\pm} = \Pi_{\pm}(\mathbf{u})\Psi$. For a uniform static magnetic field, $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$, the sectors decouple:

$$\mathrm{i}\partial_t\Psi_{\pm} = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \Psi_{\pm}. \quad (7.22)$$

For a varying field direction the sectors are coupled. With H denoting the Pauli Hamiltonian, direct differentiation gives

$$\mathrm{i}\partial_t\Psi_{\pm} = \mathrm{i}(\partial_t\Pi_{\pm})\Psi + \Pi_{\pm}H\Psi \quad (7.23)$$

$$= H\Psi_{\pm} + (\mathrm{i}\partial_t\Pi_{\pm} + [\Pi_{\pm}, H])\Psi. \quad (7.24)$$

Because the field term commutes with its own instantaneous projectors, this is

$$\mathrm{i}\partial_t\Psi_{\pm} = H\Psi_{\pm} + \mathbf{S}_{\pm}, \quad \mathbf{S}_{\pm} = \left(\mathrm{i}\partial_t\Pi_{\pm} + \frac{[\mathbf{P}^2, \Pi_{\pm}]}{2m} \right) \Psi. \quad (7.25)$$

7.5 Interacting operators and factorization

The covariant energy and momentum operators do not commute. Direct application to a test wavefunction gives

$$[E, P_m] = \mathrm{i}qE_m, \quad [P_m, P_n] = \mathrm{i}q(\partial_m A_n - \partial_n A_m), \quad \mathbf{P} \times \mathbf{P} = \mathrm{i}q\mathbf{B}. \quad (7.26)$$

Therefore

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - \mathrm{i}qF, \quad (7.27)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - \mathrm{i}qF^*, \quad (7.28)$$

$$E + \boldsymbol{\sigma} \cdot \mathbf{P} = \mathrm{i}\partial^* - q\Phi, \quad (7.29)$$

$$E - \boldsymbol{\sigma} \cdot \mathbf{P} = \mathrm{i}\partial - q\Phi^*. \quad (7.30)$$

Expanding the minimally coupled Klein–Gordon operator gives the fully explicit result

$$(\square + \mathrm{i}qS - q^2 \det(\Phi) + m^2 + 2\mathrm{i}q(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}))\Psi = 0. \quad (7.31)$$

The two orderings satisfy

$$0 = ((\partial + \mathrm{i}q\Phi^*)(\partial^* + \mathrm{i}q\Phi) + m^2)\Psi_1 = (-E^2 + \mathbf{P}^2 + m^2 + \mathrm{i}qF^*)\Psi_1, \quad (7.32)$$

$$0 = ((\partial^* + \mathrm{i}q\Phi)(\partial + \mathrm{i}q\Phi^*) + m^2)\Psi_2 = (-E^2 + \mathbf{P}^2 + m^2 + \mathrm{i}qF)\Psi_2. \quad (7.33)$$

These follow by multiplying the Dirac operator by its mass conjugate. In detail, if $\mathbf{D} = \mathbf{W}(\mathrm{i}\partial - q\Phi^*) - m\mathbf{I}$ and $\mathbf{D}' = \mathbf{W}(\mathrm{i}\partial - q\Phi^*) + m\mathbf{I}$, then

$$\mathbf{D}'\mathbf{D} = -\mathrm{diag}((\partial + \mathrm{i}q\Phi^*)(\partial^* + \mathrm{i}q\Phi) + m^2, (\partial^* + \mathrm{i}q\Phi)(\partial + \mathrm{i}q\Phi^*) + m^2). \quad (7.34)$$

For a paravector wavefunction $\psi = \psi_s + \psi_v \cdot \sigma$,

$$\partial\psi = (\partial_t\psi_s + \partial_{\mathbf{r}} \cdot \psi_v) + (\partial_t\psi_v + \partial_{\mathbf{r}}\psi_s + \mathrm{i}\partial_{\mathbf{r}} \times \psi_v) \cdot \sigma, \quad (7.35)$$

$$\partial^*\psi = (\partial_t\psi_s - \partial_{\mathbf{r}} \cdot \psi_v) + (\partial_t\psi_v - \partial_{\mathbf{r}}\psi_s - \mathrm{i}\partial_{\mathbf{r}} \times \psi_v) \cdot \sigma, \quad (7.36)$$

$$\Phi\psi = (V\psi_s + \mathbf{A} \cdot \psi_v) + (V\psi_v + \psi_s\mathbf{A} + \mathrm{i}\mathbf{A} \times \psi_v) \cdot \sigma, \quad (7.37)$$

$$\Phi^*\psi = (V\psi_s - \mathbf{A} \cdot \psi_v) + (V\psi_v - \psi_s\mathbf{A} - \mathrm{i}\mathbf{A} \times \psi_v) \cdot \sigma. \quad (7.38)$$

Writing $F = \mathbf{F} \cdot \sigma$,

$$\mathrm{sc}(\mathrm{i}qF\psi) = \mathrm{i}q\mathbf{F} \cdot \psi_v, \quad \mathrm{vec}(\mathrm{i}qF\psi) = \mathrm{i}q\psi_s\mathbf{F} - q\mathbf{F} \times \psi_v, \quad (7.39)$$

$$\mathrm{sc}(\mathrm{i}qF^*\psi) = -\mathrm{i}q\mathbf{F}^* \cdot \psi_v, \quad \mathrm{vec}(\mathrm{i}qF^*\psi) = -\mathrm{i}q\psi_s\mathbf{F}^* + q\mathbf{F}^* \times \psi_v. \quad (7.40)$$

7.6 Low-energy derivation

For spin zero, write $\Psi' = e^{-\mathrm{i}mt}\Psi$. The exact equation becomes

$$(-E^2 - 2mE + \mathbf{P}^2)\Psi = 0, \quad E = \frac{\mathbf{P}^2 - E^2}{2m} = \frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + \cdots. \quad (7.41)$$

The leading term is the Schrödinger equation

$$\mathrm{i}\partial_t\Psi = \left(\frac{(-\mathrm{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \Psi. \quad (7.42)$$

For spin one-half, the two rest-mass-stripped Dirac equations are

$$(E + m - \mathbf{P} \cdot \sigma)\Psi_2 - m\Psi_1 = 0, \quad (7.43)$$

$$(E + m + \mathbf{P} \cdot \sigma)\Psi_1 - m\Psi_2 = 0. \quad (7.44)$$

Eliminating Ψ_1 gives

$$(E^2 + 2mE - \mathbf{P}^2 + q\mathbf{B} \cdot \sigma - \mathrm{i}q\mathbf{E} \cdot \sigma)\Psi_2 = 0, \quad (7.45)$$

and iteration yields $E = (\mathbf{P}^2 - q\mathbf{B} \cdot \sigma)/(2m) + O(m^{-2})$, which is equation (7.20).

7.7 Spin, ladder operators, and energy sectors

The matrix spin projectors and their algebra counterparts are

$$\Pi_{\pm}(\mathbf{u}) := \frac{1}{2}(\mathbf{I}_2 \pm \mathbf{u} \cdot \sigma), \quad \Pi_{\pm}(\mathbf{u}) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \sigma). \quad (7.46)$$

They satisfy

$$\Pi_{\pm}(\mathbf{u})^2 = \Pi_{\pm}(\mathbf{u}), \quad \Pi_+(\mathbf{u})\Pi_-(\mathbf{u}) = 0, \quad \Pi_+(\mathbf{u}) + \Pi_-(\mathbf{u}) = \mathbf{I}_2, \quad (7.47)$$

$$\Pi_{\pm}(\mathbf{u})^{\mathrm{H}} = \Pi_{\pm}(\mathbf{u}), \quad \Pi_{\pm}(\mathbf{u})^* = \Pi_{\mp}(\mathbf{u}), \quad (\mathbf{u} \cdot \sigma)\Pi_{\pm}(\mathbf{u}) = \pm\Pi_{\pm}(\mathbf{u}). \quad (7.48)$$

Consequently

$$\mathbf{u} \cdot \boldsymbol{\sigma} = \boldsymbol{\Pi}_+(\mathbf{u}) - \boldsymbol{\Pi}_-(\mathbf{u}). \quad (7.49)$$

This left measurement projection along $\mathbf{u}$ is distinct from the fixed right ideal projection along the freely chosen reference $\mathbf{u}_0$ in Section 7.1. For the ideal matrix $\mathbf{M}_{\psi'}$ and its nonzero column $\boldsymbol{\chi}$, define

$$\mathbf{M}_{\psi', \pm} := \boldsymbol{\Pi}_{\pm}(\mathbf{u}) \mathbf{M}_{\psi'}, \quad \boldsymbol{\chi}_{\pm} := \boldsymbol{\Pi}_{\pm}(\mathbf{u}) \boldsymbol{\chi}. \quad (7.50)$$

The components are orthogonal and complete:

$$\boldsymbol{\chi} = \boldsymbol{\chi}_+ + \boldsymbol{\chi}_-, \quad \boldsymbol{\chi}_+^H \boldsymbol{\chi}_- = 0, \quad \boldsymbol{\chi}_-^H \boldsymbol{\chi}_+ = 0. \quad (7.51)$$

For a normalized state, define its real Bloch vector by

$$s_n := \frac{\boldsymbol{\chi}^H \boldsymbol{\sigma}_n \boldsymbol{\chi}}{\boldsymbol{\chi}^H \boldsymbol{\chi}}, \quad \mathbf{s} := \{s_1, s_2, s_3\}, \quad \|\mathbf{s}\|^2 = 1. \quad (7.52)$$

The probabilities for spin along and against $\mathbf{u}$ are

$$p_{\pm}(\mathbf{u}) := \frac{\boldsymbol{\chi}_{\pm}^H \boldsymbol{\chi}_{\pm}}{\boldsymbol{\chi}^H \boldsymbol{\chi}} = \frac{\boldsymbol{\chi}^H \boldsymbol{\Pi}_{\pm}(\mathbf{u}) \boldsymbol{\chi}}{\boldsymbol{\chi}^H \boldsymbol{\chi}} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (7.53)$$

For two unit directions $\mathbf{u}, \mathbf{u}'$, the product and commutator are

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (7.54)$$

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (7.55)$$

Moreover,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (7.56)$$

Define $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$. The same-side and cross-side projector identities are

$$\boldsymbol{\Pi}_{\pm}(\mathbf{u})(\mathbf{u}' \cdot \boldsymbol{\sigma})\boldsymbol{\Pi}_{\pm}(\mathbf{u}) = \pm(\mathbf{u} \cdot \mathbf{u}')\boldsymbol{\Pi}_{\pm}(\mathbf{u}), \quad (7.57)$$

$$\boldsymbol{\Pi}_{\pm}(\mathbf{u})(\mathbf{u}' \cdot \boldsymbol{\sigma})\boldsymbol{\Pi}_{\mp}(\mathbf{u}) = \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \boldsymbol{\Pi}_{\mp}(\mathbf{u}). \quad (7.58)$$

For a state aligned with $\mathbf{u}$ and $\mathbf{u} \cdot \mathbf{u}' = \cos \theta$, the measurement probabilities along $\mathbf{u}'$ are

$$p_+(\mathbf{u}') = \frac{1}{2}(1 + \mathbf{u} \cdot \mathbf{u}') = \cos^2(\theta/2), \quad p_-(\mathbf{u}') = \frac{1}{2}(1 - \mathbf{u} \cdot \mathbf{u}') = \sin^2(\theta/2). \quad (7.59)$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through ϕ about $\mathbf{n}$, then

$$\mathbf{u} \cdot \mathbf{u}' = (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos \phi. \quad (7.60)$$

Thus $\theta = \phi$ when $\mathbf{n} \cdot \mathbf{u} = 0$.

For fluctuations $d\mathbf{u}$ and $d\mathbf{u}'$, the squared determinant identity is

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = (d\mathbf{u} \cdot d\mathbf{u}')^2 + \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (7.61)$$

For orthogonal fluctuations this reduces to

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (7.62)$$

About the third axis, define

$$\mathbf{a}_\pm := \frac{1}{2}(\boldsymbol{\sigma}_1 \pm i\boldsymbol{\sigma}_2), \quad \boldsymbol{\Pi}_\pm := \frac{1}{2}(\mathbf{I}_2 \pm \boldsymbol{\sigma}_3). \quad (7.63)$$

Then

$$[\boldsymbol{\sigma}_3, \mathbf{a}_\pm] = \pm 2\mathbf{a}_\pm, \quad \mathbf{a}_\pm^2 = 0, \quad \mathbf{a}_\pm \mathbf{a}_\mp = \boldsymbol{\Pi}_\pm, \quad (7.64)$$

$$\boldsymbol{\sigma}_1 = \mathbf{a}_+ + \mathbf{a}_-, \quad \boldsymbol{\sigma}_2 = (\mathbf{a}_+ - \mathbf{a}_-)/i, \quad \boldsymbol{\sigma}_3 = \boldsymbol{\Pi}_+ - \boldsymbol{\Pi}_-. \quad (7.65)$$

Their projector action is

$$\mathbf{a}_+ \boldsymbol{\Pi}_+ = 0, \quad \mathbf{a}_+ \boldsymbol{\Pi}_- = \mathbf{a}_+, \quad \boldsymbol{\Pi}_+ \mathbf{a}_+ = \mathbf{a}_+, \quad \boldsymbol{\Pi}_- \mathbf{a}_+ = 0, \quad (7.66)$$

$$\mathbf{a}_- \boldsymbol{\Pi}_- = 0, \quad \mathbf{a}_- \boldsymbol{\Pi}_+ = \mathbf{a}_-, \quad \boldsymbol{\Pi}_- \mathbf{a}_- = \mathbf{a}_-, \quad \boldsymbol{\Pi}_+ \mathbf{a}_- = 0. \quad (7.67)$$

Finally, on the mass shell let

$$U = \frac{E + \mathbf{P} \cdot \boldsymbol{\sigma}}{m}, \quad \boldsymbol{\Lambda}_\pm = \frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)). \quad (7.68)$$

Since $\det(U) = 1$ and $\mathbf{W}(U)^2 = \mathbf{I}$, the positive- and negative-energy projectors satisfy

$$\boldsymbol{\Lambda}_\pm^2 = \boldsymbol{\Lambda}_\pm, \quad \boldsymbol{\Lambda}_+ \boldsymbol{\Lambda}_- = 0, \quad \boldsymbol{\Lambda}_+ + \boldsymbol{\Lambda}_- = \mathbf{I}. \quad (7.69)$$

Lift the spin projectors to the Dirac block space by

$$\boldsymbol{\Pi}_{j,4} := \text{diag}(\boldsymbol{\Pi}_j, \boldsymbol{\Pi}_j), \quad j = \pm. \quad (7.70)$$

When $[\boldsymbol{\Lambda}_i, \boldsymbol{\Pi}_{j,4}] = 0$, as for a compatible spin quantization axis, their products are joint energy–spin projectors and resolve the physical Dirac column selected in Section 7.1 into four sectors:

$$\boldsymbol{\Psi} = \sum_{i=\pm} \sum_{j=\pm} \boldsymbol{\Psi}_{ij}, \quad \boldsymbol{\Psi}_{ij} = \boldsymbol{\Lambda}_i \boldsymbol{\Pi}_{j,4} \boldsymbol{\Psi}. \quad (7.71)$$

Without this commutation condition the displayed sum remains a sequential decomposition, but the factors are not joint orthogonal projectors.

8 Rotations and Lorentz boosts

8.1 Rotors

A rotation through angle ϕ about unit axis $\mathbf{u}$ is generated by

$$R = \exp\left(\frac{\phi}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}\right) = \cos \frac{\phi}{2} + \mathbf{u} \cdot i\boldsymbol{\sigma} \sin \frac{\phi}{2}, \quad R^{-1} = R^H. \quad (8.1)$$

The sandwich action $X' = R^H X R$ fixes the scalar component and rotates the spatial vector according to Rodrigues' formula,

$$\mathbf{r}' = \mathbf{r} \cos \phi + (\mathbf{u} \times \mathbf{r}) \sin \phi + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \phi)\mathbf{u}. \quad (8.2)$$

The half-angle explains the spinorial sign change $R(\phi + 2\pi) = -R(\phi)$. In the physical column representation, let $\mathbf{R} := \text{mat}(R)$ and define the pure-state density matrix

$$\rho_\chi := \frac{\chi\chi^H}{\chi^H\chi} = \frac{1}{2}(\mathbf{I}_2 + \mathbf{s} \cdot \boldsymbol{\sigma}). \quad (8.3)$$

For $\chi' = \mathbf{R}\chi$,

$$\rho_{\chi'} = \mathbf{R}\rho_\chi\mathbf{R}^H, \quad \text{tr}(\rho_{\chi'} \text{mat}(Z)) = \text{tr}(\rho_\chi \mathbf{R}^H \text{mat}(Z) \mathbf{R}). \quad (8.4)$$

Thus rotating the state is dual to conjugating the observable. For a spinor $\psi_R := R\psi$, expectation values transform as

$$\psi_R^H = \psi^H R^H, \quad \langle Z \rangle_{\psi_R} = \frac{\text{sc}(\psi^H R^H Z R \psi)}{\text{sc}(\psi^H \psi)}, \quad R^H = R^{-1}. \quad (8.5)$$

For $Z = \mathbf{u} \cdot \boldsymbol{\sigma}$, this gives $Z' = R^H Z R = \mathbf{u}' \cdot \boldsymbol{\sigma}$.

8.2 Boosts

Let rapidity θ satisfy $\beta = \tanh \theta$ and $\gamma = \cosh \theta$. A boost along $\mathbf{u}$ is

$$L = \exp\left(-\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right), \quad L^H = L, \quad L^{-1} = L^*. \quad (8.6)$$

For $\psi_L := L\psi$, the boost generator is Hermitian, so

$$\psi_L^H = \psi^H L, \quad \langle Z \rangle_{\psi_L} = \frac{\text{sc}(\psi^H L Z L \psi)}{\text{sc}(\psi^H L^2 \psi)}. \quad (8.7)$$

The similarity $L^{-1}ZL$ is a different operation: it preserves $\text{sc}(Z)$ and is not the spacetime or spinor-bilinear boost used here. The spacetime paravector therefore transforms by the Hermitian congruence $X' = LX L$, which yields the components $\mathbf{r}_\parallel = (\mathbf{r} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$, $\mathbf{r}'_\parallel = (\mathbf{r}' \cdot \mathbf{u})\mathbf{u}$, and $\mathbf{r}'_\perp = \mathbf{r}' - \mathbf{r}'_\parallel$:

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (8.8)$$

$$\mathbf{r}'_\parallel = \gamma(\mathbf{r}_\parallel - \beta t \mathbf{u}), \quad \mathbf{r}'_\perp = \mathbf{r}_\perp. \quad (8.9)$$

It preserves $\det(X) = t^2 - \mathbf{r}^2$. The electromagnetic field instead obeys $F' = LFL^*$ for $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Define $\mathbf{E}_\parallel = (\mathbf{E} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}_\perp = \mathbf{E} - \mathbf{E}_\parallel$ and likewise $\mathbf{E}'_\parallel = (\mathbf{E}' \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}'_\perp = \mathbf{E}' - \mathbf{E}'_\parallel$; define the four corresponding components of $\mathbf{B}$ and $\mathbf{B}'$ by the same formulas. Then

$$\mathbf{E}'_\parallel = \mathbf{E}_\parallel, \quad \mathbf{E}'_\perp = \gamma(\mathbf{E}_\perp + \beta \mathbf{u} \times \mathbf{B}), \quad (8.10)$$

$$\mathbf{B}'_\parallel = \mathbf{B}_\parallel, \quad \mathbf{B}'_\perp = \gamma(\mathbf{B}_\perp - \beta \mathbf{u} \times \mathbf{E}). \quad (8.11)$$

8.3 Congruences, transformed bases, and non-inertial terms

Spinor bilinears use the Hermitian congruence $Y = T^H X T$. With $d\Theta = T^{-1} dT$, direct differentiation gives the ancillary identity

$$dY = T^H(dX)T + (d\Theta)^H Y + Y d\Theta. \quad (8.12)$$

For a rotation, $T^H = T^{-1}$ and $(d\Theta)^H = -d\Theta$, so this reduces to a commutator. For a Hermitian boost it does not: the two connection terms form the congruence appropriate to the spinor bilinear. Transformed algebra bases use the distinct similarity transformation. For any invertible T , let $X' = T^{-1}XT$ and $d\Theta = T^{-1}dT$. Differentiation gives the useful connection terms

$$dX' = T^{-1}(dX)T + [X', d\Theta], \quad (8.13)$$

$$d^2X' = T^{-1}(d^2X)T + 2[dX', d\Theta] + [X', d^2\Theta] + [d\Theta, [X', d\Theta]]. \quad (8.14)$$

In time-parametrized form, $\Omega = T^{-1}dT/dt$ gives

$$\frac{dX'}{dt} = T^{-1}\frac{dX}{dt}T + [X', \Omega], \quad (8.15)$$

$$\frac{d^2X'}{dt^2} = T^{-1}\frac{d^2X}{dt^2}T + 2\left[\frac{dX'}{dt}, \Omega\right] + \left[X', \frac{d\Omega}{dt}\right] + [\Omega, [X', \Omega]]. \quad (8.16)$$

The transformed generators $\sigma'_n = T^{-1}\sigma_n T$ obey the same algebra, including $(\sigma'_n)^2 = 1$ and $\sigma'_1\sigma'_2\sigma'_3 = i$, and

$$d\sigma'_n = [\sigma'_n, T^{-1}dT]. \quad (8.17)$$

If $T = f + \mathbf{g} \cdot \boldsymbol{\sigma}$, then

$$T^{-1} = \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}, \quad (8.18)$$

$$T^{-1}dT = \frac{f df - \mathbf{g} \cdot d\mathbf{g}}{f^2 - \mathbf{g}^2} + \frac{f d\mathbf{g} - df \mathbf{g} - i\mathbf{g} \times d\mathbf{g}}{f^2 - \mathbf{g}^2} \cdot \boldsymbol{\sigma}. \quad (8.19)$$

Write the vector part of Ω in the primed basis as

$$\text{vec}(\Omega) = \frac{1}{2}(\mathbf{k} + i\boldsymbol{\omega}), \quad k_n + i\omega_n = \text{sc}(2(dT/dt)T^{-1}\sigma_n). \quad (8.20)$$

Then

$$\frac{d\boldsymbol{\sigma}'}{dt} = (-\boldsymbol{\omega} + i\mathbf{k}) \times \boldsymbol{\sigma}', \quad (8.21)$$

and, for $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}'$,

$$\frac{dZ}{dt} = \frac{dS}{dt} + \left(\frac{d\mathbf{V}}{dt} - \mathbf{V} \times \boldsymbol{\omega} + i\mathbf{V} \times \mathbf{k}\right) \cdot \boldsymbol{\sigma}'. \quad (8.22)$$

For $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}'$ these are the rotational and boost-like inertial terms in the measured velocity. When $T^{-1} = \text{adj}(T)$, the transformed basis satisfies $\text{adj}(\sigma'_n) = -\sigma'_n$ and $\det(Z) = S^2 - \mathbf{V}^2$ for $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}'$.

Collinear boosts add rapidities and therefore obey

$$\beta' = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}, \quad \gamma' = \gamma_1\gamma_2(1 + \beta_1\beta_2). \quad (8.23)$$

Products reverse under inversion:

$$(LR)^{-1} = R^H L^*, \quad (RL)^{-1} = L^* R^H, \quad (R_1 R_2)^{-1} = R_2^H R_1^H, \quad (L_1 L_2)^{-1} = L_2^* L_1^*. \quad (8.24)$$

9 Spectral functions and quaternions

9.1 Functions of a unit vector

With the algebra projectors $\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$, any function defined by a power series satisfies

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(1)\Pi_+ + f(-1)\Pi_- = \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}. \quad (9.1)$$

Consequently,

$$\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} = \pm \Pi_+ \pm i\Pi_-, \quad (9.2)$$

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2i\pi(n\Pi_+ + (\frac{1}{2} + m)\Pi_-), \quad (9.3)$$

$$(z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} = \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \quad (9.4)$$

$$e^{\theta \mathbf{u} \cdot \boldsymbol{\sigma}} = \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (9.5)$$

$$e^{i\theta \mathbf{u} \cdot \boldsymbol{\sigma}} = \cos \theta + i \sin \theta \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (9.6)$$

For $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, put $\lambda_{\pm} = t \pm \|\mathbf{r}\|$ and $\mathbf{u} = \mathbf{r}/\|\mathbf{r}\|$. Then

$$X = \lambda_+ \Pi_+ + \lambda_- \Pi_-, \quad f(X) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (9.7)$$

This includes

$$X^n = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n)\mathbf{u} \cdot \boldsymbol{\sigma}, \quad (9.8)$$

$$X^{-1} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_-, \quad (9.9)$$

$$\sqrt{X} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-, \quad (9.10)$$

$$\log X = \log \lambda_+ \Pi_+ + \log \lambda_- \Pi_-. \quad (9.11)$$

For two directions,

$$\Pi_+ \Pi'_+ = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} + \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (9.12)$$

$$\Pi_+ \Pi'_- = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} - \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (9.13)$$

$$\Pi_- \Pi'_+ = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} + \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (9.14)$$

$$\Pi_- \Pi'_- = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} - \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (9.15)$$

One useful projector evaluation is

$$\frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} = e^{-2 \operatorname{atanh}(\|\mathbf{r}\|)} \Pi_+ + e^{2 \operatorname{atanh}(\|\mathbf{r}\|)} \Pi_-. \quad (9.16)$$

9.2 Quaternion subalgebra

Define

$$q_1 = -\sigma_2 \sigma_3, \quad q_2 = -\sigma_3 \sigma_1, \quad q_3 = -\sigma_1 \sigma_2, \quad \mathbf{q} = -i\boldsymbol{\sigma}. \quad (9.17)$$

Then $q_1^2 = q_2^2 = q_3^2 = q_1 q_2 q_3 = -1$. A quaternion $Q = a + \mathbf{b} \cdot \mathbf{q}$ satisfies $Q^H = a - \mathbf{b} \cdot \mathbf{q}$ and $QQ^H = a^2 + \mathbf{b}^2$; the rotation action is $\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^H$.

10 Generalized spacetime and metric form

Let

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}, \quad dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma}. \quad (10.1)$$

Writing $dX = A dt + \mathbf{B} \cdot d\mathbf{r}$ gives

$$A = \partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}, \quad \mathbf{B} = \partial_{\mathbf{r}} f + \partial_{\mathbf{r}} (\mathbf{g} \cdot \boldsymbol{\sigma}), \quad (10.2)$$

and the interval remains

$$ds^2 = dX dX^* = df^2 - d\mathbf{g}^2. \quad (10.3)$$

Define the geometric velocity and Lorentz factor by

$$\mathbf{v} = \frac{d\mathbf{g}}{df}, \quad \gamma = \frac{df}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}, \quad U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (10.4)$$

With geometric acceleration $\mathbf{a} = d\mathbf{v}/ds = \gamma d^2\mathbf{g}/df^2$, the proper-acceleration formula is unchanged. Inertial motion $dU/ds = 0$ implies $\mathbf{a} = 0$, hence $\mathbf{g} = c_1 \mathbf{f} + c_2$ and constant geometric velocity. If $f' = df/dt$ and $\mathbf{g}' = d\mathbf{g}/dt$, then

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{g}'' f' - \mathbf{g}' f''}{(f')^2}, \quad [U, dU/ds] = 2\gamma^3 (\mathbf{v} \times \mathbf{a}) \cdot \mathbf{i}\boldsymbol{\sigma}. \quad (10.5)$$

In coordinates $\mathbf{x} = \{t, r_1, r_2, r_3\}^\top$, write

$$ds^2 = d\mathbf{x}^\top \mathbf{Q}(\mathbf{x}) d\mathbf{x}. \quad (10.6)$$

For real symmetric $\mathbf{Q} = \mathbf{N} \mathbf{D} \mathbf{N}^\top$ with eigenvalues λ_n and orthonormal eigenvectors $\mathbf{u}_n$,

$$ds^2 = \sum_n \lambda_n (\mathbf{u}_n \cdot d\mathbf{x})^2, \quad dx'_n = \sqrt{\lambda_n} \mathbf{u}_n \cdot d\mathbf{x}, \quad \mathbf{Q}^{-1} = \mathbf{N} \mathbf{D}^{-1} \mathbf{N}^\top. \quad (10.7)$$

11 Reference summary

This index collects the principal definitions and results in natural units, with signature $(+, -, -, -)$. Topic titles and equation numbers link to their full statements and derivations.

Algebra and vector operations.

$$\begin{aligned} \sigma_n \sigma_m &= \delta_{nm} + \epsilon_{nmk} \mathbf{i} \sigma_k, & [\sigma_n, \sigma_m] &= 2\epsilon_{nmk} \mathbf{i} \sigma_k & (1.10), \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} + \mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, & \mathbf{A}^2 &= \mathbf{A} \cdot \mathbf{A}, \\ \|Z\|^2 &= \|S\|^2 + \|\mathbf{V}\|^2, & \|\mathbf{V}\|^2 &= \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} & (1.5). \end{aligned}$$

Involutions and adjugation.

$$\begin{aligned} (ZZ')^* &= Z^* Z'^*, & (ZZ')^H &= Z'^H Z^H, \\ \text{adj}(Z) &= Z^{*H}, & \text{adj}(ZZ') &= Z'^{*H} Z^{*H} & (2.1). \end{aligned}$$

Paravectors, extraction, and products. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$,

$$\begin{aligned} S &= \text{sc}(Z) = \frac{1}{2}(Z + Z^{*\text{H}}), & \mathbf{V} \cdot \boldsymbol{\sigma} &= \frac{1}{2}(Z - Z^{*\text{H}}), \\ a &= \text{sc}(\text{re}(Z)), & b &= \text{sc}(\text{im}(Z)), \\ \mathbf{A} &= \text{vec}(\text{re}(Z)), & \mathbf{B} &= \text{vec}(\text{im}(Z)), \\ ZZ' &= (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}, \\ \det(Z) &= \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2, & Z^{-1} &= \frac{\text{adj}(Z)}{\det(Z)}, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0, & \lambda_{\pm} &= S \pm \sqrt{\mathbf{V}^2}. \end{aligned}$$

See the product (3.15), determinant (3.18), and quadratic identity (3.22).

Pauli-matrix representation.

$$\begin{aligned} \text{mat}(Z) &= \begin{pmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{pmatrix}, & \text{tr}(Z) &= 2S \\ \mathbf{M}^{\text{H}} &= (\mathbf{M}^{\text{C}})^{\text{T}} = (\mathbf{M}^{\text{T}})^{\text{C}}, & \mathbf{M}^* &= -\mathbf{J}\mathbf{M}^{\text{C}}\mathbf{J}, \\ \text{mat}(\text{adj}(Z)) &= \text{adj}(\mathbf{M}), & \text{adj}(\mathbf{M}) &= \text{tr}(\mathbf{M})\mathbf{I}_2 - \mathbf{M}. \end{aligned} \tag{4.1}$$

Space-time kinematics and action.

$$\begin{aligned} dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, & ds^2 &= dX dX^* = dt^2 - d\mathbf{r}^2 \\ U &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), & \gamma &= (1 - \mathbf{v}^2)^{-1/2}, \\ mU &= E + \mathbf{P} \cdot \boldsymbol{\sigma}, & E^2 - \mathbf{P}^2 &= m^2 \\ S &= -m \int ds = \int L dt, & L &= -m/\gamma, \quad H = m\gamma. \end{aligned} \tag{5.2}$$

$$\tag{5.6}$$

Electromagnetism.

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, & F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, & D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma}, \\ \partial F &= D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}, & \square &= \partial^* \partial & (6.2), \\ \partial \Phi &= S + F^*, & \mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, & \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}, \\ \frac{1}{2}F^2 &= \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, & \frac{1}{2}FF^{\text{H}} &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ m \frac{dU}{ds} &= q \text{re}(FU), & \frac{d\mathbf{P}}{dt} &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) & (6.11). \end{aligned}$$

Wave equations, spin, and energy sectors.

$$\begin{aligned}
\langle \phi | \psi \rangle &= \text{sc}(\phi^H \psi) = \frac{1}{2} \text{tr}(\mathbf{M}_\phi^H \mathbf{M}_\psi), \\
\langle Z \rangle_\psi &= \frac{\text{sc}(\psi^H Z \psi)}{\text{sc}(\psi^H \psi)}, \\
(\square + m^2)\Psi &= 0, \\
\mathbf{W}(Z) &= \begin{pmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{pmatrix}, \\
(\mathbf{W}(\text{i}\partial) - m\mathbf{I})\Psi &= 0, \\
(\mathbf{W}(\text{i}\partial - q\Phi^*) - m\mathbf{I})\Psi &= 0, \\
\text{i}\partial_t \Psi &= \left(\frac{(-\text{i}\partial_r - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \Psi, \\
\Pi_\pm(\mathbf{u}) &= \frac{1}{2}(\mathbf{I}_2 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
p_\pm(\mathbf{u}) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}), \\
\Lambda_\pm &= \frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)), \quad \Pi_{j,4} = \text{diag}(\Pi_j, \Pi_j), \\
\Psi &= \sum_{i=\pm} \sum_{j=\pm} \Psi_{ij}, \quad \Psi_{ij} = \Lambda_i \Pi_{j,4} \Psi.
\end{aligned}$$

See the paravector inner product (7.1), minimal-ideal spinor (7.9), Klein–Gordon equation (7.12), Dirac equations (7.17) and (7.19), Pauli equation (7.20), and spin probability (7.53).

Rotations, boosts, and transformed bases.

$$\begin{aligned}
R &= e^{\phi \mathbf{u} \cdot \mathbf{i}\boldsymbol{\sigma}/2}, \quad R^H = R^{-1}, \quad Z' = R^H Z R, \\
L &= e^{-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2}, \quad L^H = L, \quad X' = L X L, \\
\langle Z \rangle_{\psi_T} &= \frac{\text{sc}(\psi^H T^H Z T \psi)}{\text{sc}(\psi^H T^H T \psi)}, \\
\text{d}(T^H X T) &= T^H(\text{d}X)T + (\text{d}\Theta)^H T^H X T + T^H X T \text{d}\Theta, \\
X' &= T^{-1} X T, \\
\text{d}X' &= T^{-1}(\text{d}X)T + [X', T^{-1} \text{d}T].
\end{aligned}$$

See the bilinear rule (7.11) and congruence derivative (8.12).

Spectral functions and quaternions.

$$\begin{aligned}
f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\Pi_+ + f(-1)\Pi_-, \\
X &= \lambda_+ \Pi_+ + \lambda_- \Pi_-, \quad f(X) = f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_-, \\
\mathbf{q} &= -\text{i}\boldsymbol{\sigma}, \quad Q^H = a - \mathbf{b} \cdot \mathbf{q}, \quad QQ^H = a^2 + \mathbf{b}^2.
\end{aligned}$$

Generalized spacetime and metric form.

$$\begin{aligned}
X &= f + \mathbf{g} \cdot \boldsymbol{\sigma}, & \text{d}s^2 &= \text{d}f^2 - \text{d}\mathbf{g}^2, \\
\mathbf{v} &= \text{d}\mathbf{g}/\text{d}f, & U &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\
\text{d}s^2 &= \text{d}\mathbf{x}^\text{T} \mathbf{Q}(\mathbf{x}) \text{d}\mathbf{x}, & \mathbf{Q} &= \mathbf{N} \mathbf{D} \mathbf{N}^\text{T}.
\end{aligned}$$

Conventions and cautions

- The same symbol $i = \sigma_1\sigma_2\sigma_3$ is both the algebra’s pseudoscalar and, in the Pauli representation, the ordinary complex unit.
- Sigma conjugation $(\cdot)^*$, reversion $(\cdot)^H$, and their composition are distinct. In the Pauli representation, reversion corresponds to Hermitian adjunction.
- $\det(Z)$ is an indefinite quadratic form and may be complex for complex Z .
- Differential operators act on everything to their right unless parentheses state otherwise. This matters in products involving fields.

*Editorial notation record.*¹

¹Compared with `tmp.txt`, the following presentation and style changes were made. Vectors and matrices are bold, and only vectors and matrices are bold; in particular, the electromagnetic fields formerly written as calligraphic or italic E and B are the upright bold vectors $\mathbf{E}$ and $\mathbf{B}$, while the scalar energy remains the unbold symbol E . Ordinary capital scalar quantities, including the action S , Lagrangian L , and Hamiltonian H , use the regular math font rather than calligraphic or other decorative alphabets. Blackboard bold is retained only for the number sets $\mathbb{R}$ and $\mathbb{C}$. The Dirac block function is the upright bold $\mathbf{W}(\cdot)$. A vector square is explicitly defined by $\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}$. Curly braces are reserved for explicit vector-entry lists, and commutators alone use square brackets. Parentheses delimit matrices, functions, algebraic grouping, and order-sensitive operator products; redundant grouping parentheses were removed. The extraction functions $\text{sc}(\cdot)$, $\text{vec}(\cdot)$, $\text{re}(\cdot)$, and $\text{im}(\cdot)$ replace `scalar`, `vector`, and capitalized variants and are defined before use; the unused expectation abbreviation `Exp` was removed. The original metric notation $\langle\langle Z \rangle\rangle^2$ and every determinant bar were replaced by the unambiguous function notation $\det(Z)$. Double bars are reserved for squared scalar, vector, and coefficient norms, with $\|Z\|^2 = \|S\|^2 + \|\mathbf{V}\|^2$; absolute-value bars are not used. Entrywise complex-conjugation notation is confined to the matrix section. Sigma conjugation retains the splat `*`; reversal, originally R and elsewhere a dagger, is written H , because it reverses the multiplied sigma order and becomes Hermitian adjunction in the matrix representation. The abbreviation $\text{adj}(Z) := Z^{*H}$ is defined early and used where it shortens formulas. The matrix section uses matching upright sans-serif capitals T , C , and H for transpose, entrywise complex conjugation, and Hermitian transpose, and distinguishes these from sigma conjugation, including which operations reverse product order. Plus/minus labels are subscripts ($\pm$, $+$, $-$), not superscripts or parenthesized labels. Parallel and perpendicular components use $\parallel$ and $\perp$ rather than `par` and `prp`, with their projection formulas restated locally at each use. Spatial derivatives use ∂_r rather than ∇ , and no overbars are used. Literal or malformed source spacing and fraction commands such as `quad` and `frac` were converted to properly formatted L^AT_EX. Long product expansions were consolidated while retaining their distinct components and results, and expressions otherwise preserve the compact ordering of the source, for example $\text{sc}(\Phi^*U)$ rather than adding an extra pair of parentheses.